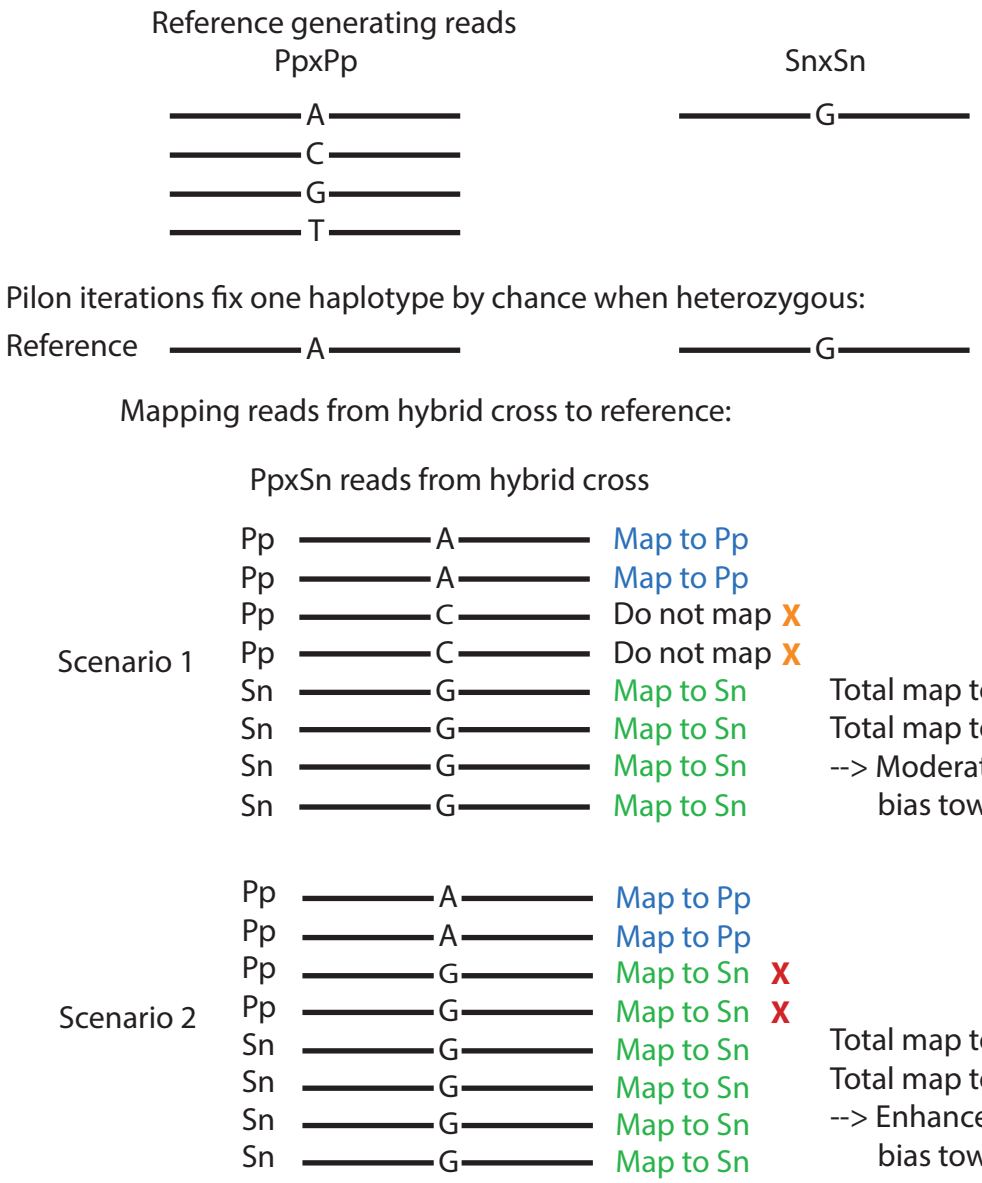

Supplementary Figure 3. Investigating the mapping behavior of sequencing reads derived from heterozygous populations mapped to a fixed reference transcriptome. Intrapopulation crosses were sequenced to generate reads to make two new population specific references, designated as Reference PpxPp and Reference SnxSn. At a particular single nucleotide polymorphism (SNP) position, Reference PpxPp is known/postulated to be heterozygous with (two to?) four possible haplotypes (A, C, G, or T). However, due to the randomness in reference selection during Pilon iterations, the reference transcriptome for PpxPp used in the final analysis represents the allele 'A' at this SNP position. Conversely, Reference SnxSn is homozygous at the same SNP position, consistently showing the allele 'G'. The sequencing reads utilized in this experiment are derived from a hybrid of the two references, containing a mix of alleles from both Reference Pp and Reference Sn. In this figure two separate scenarios are presented that will lead to a map bias towards the SnxSn reference to a varrying degree, while the expected map ratio should be 50/50 in the hybrid.